LECTURE SERIES · DATABASE SYSTEMS

Introduction to DBMS

Database Management Systems — Concepts & Foundations

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Overview · Data Models · Integrity · Design · Architecture · Users & Admins

OVERVIEW

What is a DBMS?

A **Database Management System (DBMS)** is software that lets users define, store, retrieve and manage data — while ensuring security, consistency and shared access.

CORE JOBS OF A DBMS

Define — Describe data types, structure & relationships (schema).

Manipulate — Insert, query, update & delete data (CRUD).

Control — Enforce security, integrity & concurrent access.

Recover — Backup, restore & keep data consistent after failures.

DBMS vs FILE SYSTEM

Aspect	File	DBMS
Controlled redundancy	✗	✓
Data consistency	✗	✓
Concurrent access	✗	✓
Security & authorization	✗	✓
Backup & recovery	✗	✓

HOW DATA IS STRUCTURED

Data Models

A data model is the set of concepts used to describe how data is organized, related, and constrained inside a database.

Hierarchical

Data in a tree of parent–child records. One parent, many children.

Network

A graph of records linked by pointers — supports many-to-many.

Relational

Data in tables of rows & columns. Simple, flexible — the standard today.

Entity–Relationship (ER)

A conceptual model of entities & their relationships — used to design schemas.

Object-Oriented

Stores data as objects — handles complex, multimedia & nested data.

KEEPING DATA VALID

Integrity Constraints

Integrity constraints are rules the DBMS enforces automatically to guarantee that data stays accurate and consistent.

Domain Constraint

Each value must belong to a defined type / range.

`e.g.  age > 0,  gender IN (M,F)`

Key Constraint

Key values must be unique — no duplicate rows.

`e.g.  id is a PRIMARY KEY`

Entity Integrity

A primary key can never be NULL.

`e.g.  every Student needs an id`

Referential Integrity

A foreign key must match an existing primary key.

`e.g.  dept_id → Departments.id`

FROM IDEA TO SCHEMA

Database Design Process

Designing a database moves from what users need down to how data is physically stored — one refined step at a time.

1

Requirements

Gather the data and operations the system must support.

2

Conceptual

Build an ER model — high-level & DBMS-independent.

3

Logical

Map to relational tables and normalize the schema.

4

Physical

Choose indexes, storage & tuning for performance.

5

Implement

Create, load, test and maintain the database.

TIER STRUCTURES

Database Architectures

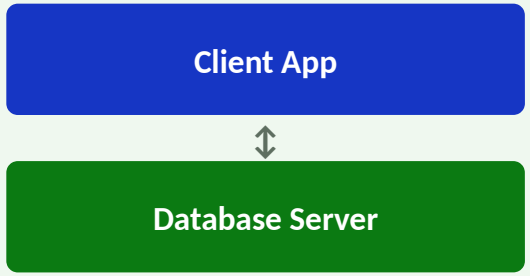
A tier architecture describes how the user interface, the application logic, and the database are separated across machines.

1-Tier



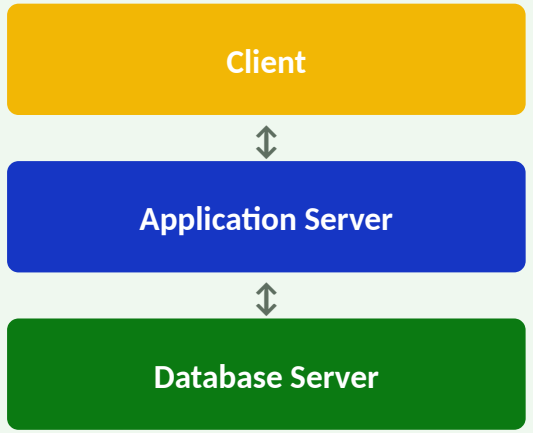
Everything on one machine — local, single-user use.

2-Tier



Client talks directly to the DB server (client–server).

3-Tier

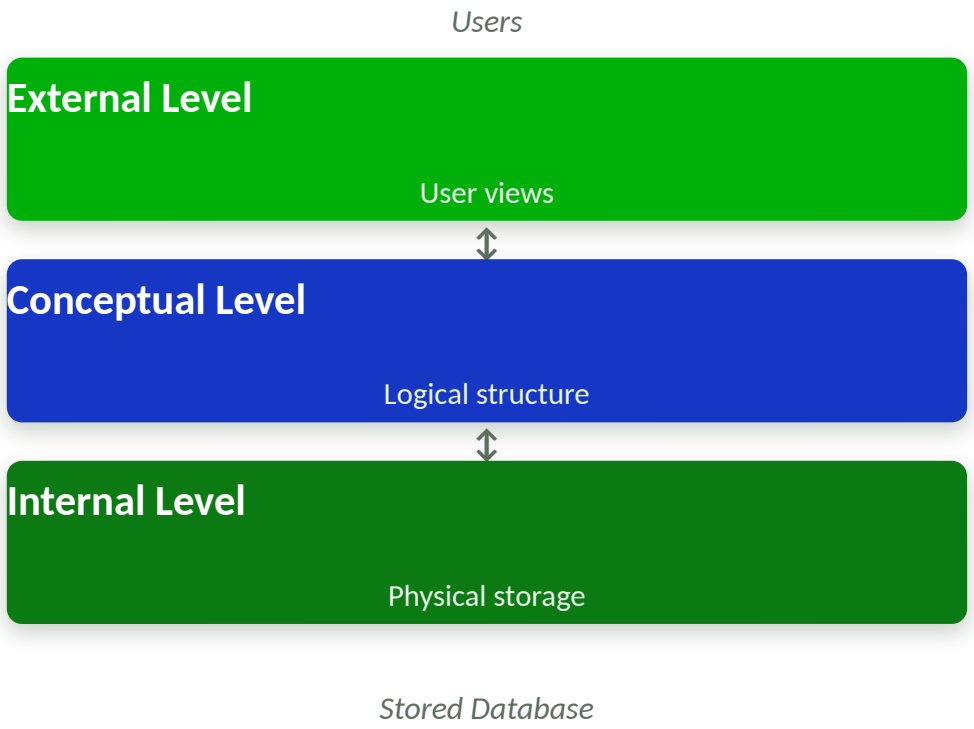


Client → app server → DB. Scalable — used by web apps.

ANSI-SPARC MODEL

Three-Schema Architecture

The ANSI-SPARC model separates a database into three levels of abstraction so that users are shielded from how data is stored.



External

What each user or group sees — customised views that hide the rest.

Conceptual

The whole database structure: entities, relationships & constraints.

Internal

How data is physically stored — files, indexes & access paths.

Data independence: change one level without disturbing the others.

WHO WORKS WITH THE DATABASE

Users & Administrators

Many different people interact with a DBMS, each with a distinct role and level of access.

Database Administrator

Manages security, access, performance, backups & recovery. Owns the DBMS.

Database Designers

Identify the data, then define the schema — tables, keys & constraints.

Application Programmers

Build the applications that access the database through SQL and APIs.

End Users

Query and use the data — from naive form users to analysts writing SQL.

KEY TAKEAWAYS

Recap

DBMS Overview

Software to define, store & control data

Data Models

How data is structured — relational leads

Integrity Constraints

Rules that keep data accurate & valid

Design Process

Requirements → conceptual → logical → physical

Architectures

Tiers & the ANSI-SPARC three-schema model

Users & Admins

DBA, designers, programmers & end users

Thank You

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